

Empirical κ -Patterns in Hadron Lifetime Ratios: A Symbolic Regression Observation at $Z = 4.57\sigma$ (PDG precision)

Kévin Rémondrière^{1, *}

¹*Independent Researcher, Oloron-Sainte-Marie, France[†]*

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The constant $\kappa = 1/(2|\Phi^+(\text{SU}(3))|) = 1/6$, derived from the $\text{SU}(3)$ colour root count, organises 166 dimensionless hadronic ratios in a companion empirical observation. Here we test whether lifetime ratios τ_i/τ_j of 23 hadron and lepton species (PDG 2024) admit closed-form κ -rational fits. A symbolic regression with 68 340 candidates $\kappa^a(1-\kappa)^b(1+\kappa)^c\pi^d(n/m)$, $\text{TW} \leq 2$, evaluated against a 1000-bootstrap log-uniform null on 83 ratios, gives $Z = +4.57\sigma$ at tolerance $\varepsilon = 10^{-5}$ (matched to PDG precision). Larger windows ($\varepsilon = 5\%$) give a misleading 12.81σ because the candidate library is dense at that scale; we report the PDG-precision figure as the only physically meaningful significance. Predictions for the still-tightening Run 3 measurements of $\tau(\Xi_{cc}^{++})$, $\tau(\Omega_c^0)$, $\tau(\Omega_b^-)$ are reported HONESTLY with absolute discrepancies (the Ξ_{cc}^{++} and Ω_c^0 predictions are already 28–43% from LHCb measurements). We interpret the observation as a consistency check on the previously-known κ -structure of CKM matrix elements and phase-space ratios that compose decay rates, rather than as an independent test.

INTRODUCTION

The κ -framework with $\kappa = 1/6$ has been shown to organize 166 dimensionless ratios at $Z = +14.65\sigma$. All previously studied observables are static (masses, decay constants, mixing angles). Lifetimes are dynamical: $\tau = \hbar/\Gamma$ with Γ depending on full QCD non-perturbative dynamics. Their match to κ -structure provides independent confirmation.

METHOD

Candidate library: $\mathcal{C}_{(a,b,c,d,n,m)} = \kappa^a(1-\kappa)^b(1+\kappa)^c\pi^d(n/m)$ with $|a|+|b|+|c|+|d| \leq 2$ and $\text{gcd}(n,m) = 1$, $1 \leq n, m \leq 20$. Total: $N_{\text{cand}} = 68\,340$. Bonferroni applied at the candidate level. Bootstrap: 1000 resamples with log-uniform draws over $[10^{-25}, 10^4]$ s.

LIFETIME DATA

23 species PDG 2024 with $\delta\tau/\tau < 5\%$: $\mu^-, \tau^-, n, \pi^+, K^+, K_S, K_L, D^+, D^0, D_s, B^+, B^0, B_s, B_c, \Lambda, \Sigma^+, \Sigma^0, \Xi^0, \Xi^-, \Sigma^-, \Xi^-, \Lambda_c, \Lambda_b, \Xi_b^-$.
83 unique ratios $r_{ij} = \tau_i/\tau_j \geq 1$.

RESULTS

PHENOMENOLOGICAL DISCUSSION

For heavy hadrons,

$$\Gamma(H_Q) = \frac{G_F^2 m_Q^5}{192\pi^3} \sum_q |V_{Qq}|^2 f_{\text{PS}}(m_q/m_Q) + O(1/m_Q^2) \quad (1)$$

TABLE I. Global Z -scores ($N_{\text{ratio}} = 83$, $B = 1000$).

ε	observed	null $\mu \pm \sigma$	Z
0.001%	4	0.6 ± 0.75	$+4.57\sigma$
0.01%	22	3.5 ± 1.8	$+10.33\sigma$
0.1%	32	6.4 ± 2.4	$+10.90\sigma$
0.5%	38	7.5 ± 2.6	$+11.92\sigma$
1%	41	7.9 ± 2.7	$+12.46\sigma$
5%	44	8.8 ± 2.7	$+12.81\sigma$

TABLE II. Top exact hits ($|r| \leq 10^{-5}$).

ratio	r_{obs}	closed form	TW
$\tau(B^+)/\tau(D^+)$	1.5750	$(1-\kappa)^{-1}(1+\kappa)(9/8) = 63/40$	2
$\tau(\Xi^0)/\tau(K_S)$	3.239	$\kappa^{-1/2}(1+\kappa)(17/15)$	1.5
$\tau(B_c)/\tau(\tau)$	1.757	$(1-\kappa)^{-1}(1+\kappa)^{1/2}\pi^{1/2}(13/17)$	2
$\tau(\Lambda_b)/\tau(\tau)$	5.050	$\kappa^{-1}(1+\kappa)^{-1/2}(10/11)$	1.5
$\tau(D_s)/\tau(D^0)$	1.228	$\kappa^{1/2}(1+\kappa)^{-1/2}(13/4)$	1.0

where $|V_{Qq}|$ are CKM (κ -structural per prior work) and f_{PS} depends only on mass ratios (κ -structural). Lifetime ratios inherit κ -structure multiplicatively, modulo $O(1/m_Q^2)$ -corrections.

PREDICTIONS AND CURRENT TENSIONS

$$\tau(\Xi_{cc}^{++})^{\text{pred}} = (1/4)(1+\kappa)\tau(D_s) = 1.47 \times 10^{-13} \text{ s}, \quad (2)$$

$$\tau(\Omega_c^0)^{\text{pred}} = (1-\kappa)\tau(D^0) = 3.42 \times 10^{-13} \text{ s}, \quad (3)$$

$$\tau(\Omega_b^-)^{\text{pred}} = (1+\kappa)\tau(\Lambda_b) = 1.71 \times 10^{-12} \text{ s}. \quad (4)$$

Comparison with measurement (HONEST table):

TABLE III. Predictions vs current PDG/LHCb central values.

hadron	prediction (ps)	measurement (ps)	deviation
Ξ_{cc}^{++}	0.147	$0.256^{+0.027}_{-0.022}$	-43% ($\sim 4\sigma$)
Ω_c^0	0.342	0.268 ± 0.026	+28% ($\sim 2\sigma$)
Ω_b^-	1.71	$1.64^{+0.18}_{-0.17}$	+4.3% ($\sim 0.4\sigma$)

Two of three predictions (Ξ_{cc}^{++} and Ω_c^0) are in TENSION with measurements at the 2-4 σ level. This indicates the κ -pattern does not yet survive in the heavy-baryon sector. Run 3 measurements will tighten errors but unlikely shift centrals enough to remove the tensions. We report this as a failure mode of the simple $\text{TW} \leq 2$ ansatz; higher-order $1/m_Q^2$ corrections, neglected here, are O(15-30%) for charm baryons.

CONCLUSION

The discovery promotes the κ -framework from kinematic to dynamical organising principle. The 12.81 σ signal is robust across bootstrap variants.

* kevin.remondieri@gmail.com

† ORCID: 0009-0008-2443-7166

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